

Auteur: Rob Willemsen
 Studierichting: Informatica
 Oefeningenreeks 2D nr. 11.9

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \left(\frac{\sqrt{2+x} - \sqrt{2-x}}{x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(\sqrt{2+x} - \sqrt{2-x})(\sqrt{2+x} + \sqrt{2-x})}{x(\sqrt{2+x} + \sqrt{2-x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sqrt{2+x}^2 - \sqrt{2-x}^2}{x(\sqrt{2+x} + \sqrt{2-x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{(2+x) - (2-x)}{x(\sqrt{2+x} + \sqrt{2-x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{2x}{x(\sqrt{2+x} + \sqrt{2-x})} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{2}{\sqrt{2+x} + \sqrt{2-x}} \right) \\
 &= \frac{2}{\sqrt{2+0} + \sqrt{2-0}} \\
 &= \frac{2}{\sqrt{2} + \sqrt{2}} \\
 &= \frac{2}{2\sqrt{2}} \\
 &= \frac{1}{\sqrt{2}}
 \end{aligned}$$

References